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Poultry processing apparatus

Abstract

A poultry processing apparatus according to an embodiment includes: a scraper part for separating a meat part around scapula bones of a poultry carcass held by a holder from the scapula bones; and an incision blade with a cutting edge for incising a skin between the scapula bones of the meat part that rides up on the scraper part. The cutting edge extends in a direction away from the holder as the cutting edge extends upstream in a traveling direction in which the scraper part travels toward the holder in separating the meat part from the scapula bones.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5098337	12/1991	Landt	452/136	A22C 21/0069
5984770	12/1998	Asano et al.	N/A	N/A
6007416	12/1998	Janssen	452/136	A22C 21/003
6277020	12/2000	Stephens	452/169	A22C 21/0023
9706782	12/2016	Inoue et al.	N/A	N/A
2009/0170417	12/2008	Janssen	452/136	A22C 21/0053
2017/0119002	12/2016	Inoue et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H1033110	12/1997	JP	N/A
2011125317	12/2010	JP	N/A
6216056	12/2016	JP	N/A
2016002629	12/2015	WO	N/A

OTHER PUBLICATIONS

International Preliminary Report on Patentability issued in Intl. Appln. No. PCT/JP2023/026498, mailed Feb. 13, 2025. English translation provided. cited by applicant
English translation of Written Opinion issued in Intl. Appln. No. PCT/JP2023/026498, mailed Aug. 22, 2023, Jun. 4, 2024. cited by applicant
International Search Report issued in Intl. Appln. No. PCT/JP2023/026498, mailed Aug. 22, 2023. English translation provided. cited by applicant
Written Opinion issued in Intl. Appln. No. PCT/JP2023/026498, mailed Aug. 22, 2023. cited by applicant

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a poultry processing apparatus for cutting up poultry carcasses of fowls such as chickens.

BACKGROUND

(2) Generally, in the cut-up processing of cutting up poultry carcasses of fowls such as chickens and separating meat and bone, the carcasses are plucked and bled, and pre-processed by removing guts and other organs (evisceration), followed by cutting and deboning. Since the cutting and deboning process is inefficient when performed manually, automation is being promoted (see, for example, Patent Document 1).

CITATION LIST

Patent Literature

(3) Patent Document 1: JP6216056B

SUMMARY

Problems to be Solved

(4) The scapula incision-making apparatus described in the above-mentioned patent document can separate a meat part around scapula bones of a poultry carcass from the scapula bones. After the meat part around the scapula bones is separated from the scapula bones, the skin between the scapula bones is incised from the outside of the meat part through the process of incising the skin between the scapula bones. In this process, there is a risk of damaging the neck meat part of the poultry carcass. Cutting and removing the damaged portion of the neck meat part may cause the remaining portion of the neck meat part to split wide open and reduce the commercial value of the neck meat part.

(5) In view of the above, an object of at least one embodiment of the present disclosure is to provide a poultry processing apparatus that can improve the yield of neck meat part of poultry carcasses.

Solution to the Problems

(6) (1) A poultry processing apparatus according to at least one embodiment of the present disclosure includes: a scraper part for separating a meat part around scapula bones of a poultry carcass held by a holder from the scapula bones; a cutting edge extending in a direction away from the holder as the cutting edge extends upstream in a traveling direction in which the scraper part travels toward the holder in separating the meat part from the scapula bones; and an incision blade with the cutting edge for incising a skin between the scapula bones of the meat part that rides up on the scraper part.

Advantageous Effects

(7) According to at least one embodiment of the present disclosure, it is possible to improve the yield and quality of neck meat part of poultry carcasses.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an overall schematic diagram of a poultry processing apparatus according to an embodiment.

(2) FIG. 2 is a perspective view of an incision making part according to an embodiment.

- (3) FIG. 3 is a cross-sectional view taken along line A-A in FIG. 2.
- (4) FIG. 4A is a perspective view of a pair of scraper members.
- (5) FIG. 4B is a perspective view of the pair of scraper members when viewed from a different direction than FIG. 4A.
- (6) FIG. 5 is a schematic cross-sectional view showing the separation of a meat part around scapula bones from the scapula bones and the incision of a skin between the scapula bones with the incision making part.
- (7) FIG. 6 is a cross-sectional view taken along line B-B in FIG. 2.
- (8) FIG. 7A is a diagram for describing the operation of a posture adjustment device to raise a neck transition part.
- (9) FIG. 7B is a diagram for describing the operation of a posture adjustment device to raise a neck transition part.
- (10) FIG. 7C is a diagram for describing the operation of a posture adjustment device to raise a neck transition part.
- (11) FIG. 7D is a diagram for describing the operation of a posture adjustment device to raise a neck transition part.

DETAILED DESCRIPTION

- (12) Embodiments of the present disclosure will be described below with reference to the accompanying drawings. It is intended, however, that unless particularly identified, dimensions, materials, shapes, relative positions, and the like of components described in the embodiments shall be interpreted as illustrative only and not intended to limit the scope of the present disclosure.
- (13) For instance, an expression of relative or absolute arrangement such as “in a direction”, “along a direction”, “parallel”, “orthogonal”, “centered”, “concentric” and “coaxial” shall not be construed as indicating only the arrangement in a strict literal sense, but also includes a state where the arrangement is relatively displaced by a tolerance, or by an angle or a distance whereby it is possible to achieve the same function.
- (14) For instance, an expression of an equal state such as “same” “equal” and “uniform” shall not be construed as indicating only the state in which the feature is strictly equal, but also includes a state in which there is a tolerance or a difference that can still achieve the same function.
- (15) Further, for instance, an expression of a shape such as a rectangular shape or a cylindrical shape shall not be construed as only the geometrically strict shape, but also includes a shape with unevenness or chamfered corners within the range in which the same effect can be achieved.
- (16) On the other hand, an expression such as “comprise”, “include”, “have”, “contain” and “constitute” are not intended to be exclusive of other components.
- (17) FIG. 1 is an overall schematic diagram of a poultry processing apparatus according to an embodiment.
- (18) FIG. 2 is a perspective view of an incision making part according to an embodiment.
- (19) FIG. 3 is a cross-sectional view taken along line A-A in FIG. 2.
- (20) The poultry processing apparatus **10** is equipped with an endless chain conveyor **12** arranged in the horizontal direction. The chain conveyor **12** is wound around a driving sprocket **14** at the beginning. The driving sprocket **14** is driven by a motor **16**. In the figures, the chain conveyor **12** moves in the direction of arrow a during the outward journey. The endless chain conveyor **12** is wound around a driven sprocket (not shown) at the end.
- (21) The chain conveyor **12** is equipped with a plurality of fixing jigs **20** called “cones” at equal intervals. In the illustrated embodiment, the upper half of the fixing jig **20** has a conical shape and stands perpendicular to the chain conveyor **12**, but it is tiltable as needed during the conveyance.
- (22) The chain conveyor **12** forms a conveyance path for the fixing jig **20**, and the fixing jig **20** is conveyed along the conveyance path.
- (23) A poultry carcass (hereinafter, referred to as “workpiece”) w of a fowl such as a chicken, with legs separated and eviscerated to leave only the upper half in pre-processing, is placed and fixed on

each fixing jig **20** by a worker at the beginning of the chain conveyor **12**. The workpiece **w** is placed and held on the fixing jig **20** with a breast part facing upstream or downstream in the conveying direction (in the illustrated embodiment, downstream in the conveying direction). The fixing jig **20** is also referred to as a holder.

(24) In the exemplary embodiment, the apparatus is equipped with a conveyance distance detecting unit to detect the conveyance distance of the fixing jig **20** from a reference point of the chain conveyor **12**.

(25) The conveyance distance detecting unit is composed of, for example, an encoder **18** installed in the motor **16** for detecting the cumulative number of revolutions of the motor **16**. By measuring the cumulative number of revolutions of the motor **16** with the encoder **18**, for example, the amount of movement (conveyance distance) of each fixing jig **20** from the motor position can be detected using the motor position as the reference point.

(26) In some embodiments, as shown in FIGS. **2**, **8**, and **10**, an incision making part **22** is provided above the chain conveyor **12**.

(27) The incision making part **22** includes a scraper part **24** vertically movable above the chain conveyor **12**, a drive device for vertically moving the scraper part **24**, an incision blade **60** attached to the scraper part **24**, and a posture adjustment device **40**.

(28) The drive device is composed of, for example, a servo motor **26** supported by a support **28**, as shown in FIG. **2**. The servo motor **26** rotates a screw shaft **26a** that passes vertically through the support **28**. The screw shaft **26a** is provided with a movable block **30** that is slidable on the screw shaft **26a** by rotation of the screw shaft **26a**. A bracket **32** is rotatably attached to the movable block **30** at one end via a shaft **32a**, and the scraper part **24** hangs from the other end.

(29) In the exemplary configuration, the incision making part **22** includes an elastic support part which elastically supports the scraper part **24** and the incision blade **60** attached to the scraper part **24** so that they can retract downstream in the conveying direction of the workpiece **w**.

(30) The elastic support part is composed of an air cylinder **34** fixed to the support **28**. A piston rod **34a** of the air cylinder **34** passes downward through the support **28** and is coupled to the end of the bracket **32** on the side where the scraper part **24** is fixed. This allows the scraper part **24** to escape downstream in the conveying direction (direction of arrow **b**) when a reaction force above a certain level is applied to the scraper part **24** from the workpiece **w**.

(31) Further, the incision making part **22** is configured to move the scraper part **24** along the scapula bone surface of the workpiece **w** by operating the servo motor **26** to lower the scraper part **24** and the incision blade **60** attached to the scraper part **24** in accordance with the timing when the workpiece **w** fixed to the fixing jig **20** comes to the meat separation position below the scraper part **24**. In the exemplary embodiment, the apparatus is equipped with a synchronizing device (not shown) that enables this operation of the scraper part **24** based on a detection value detected by the encoder **18**.

(32) With this configuration, the scraper part **24** separates a meat part around the scapula bones of the workpiece **w** that has come to the meat separation position from the scapula bones, and the incision blade **60** incises a skin between the scapula bones. The scraper part **24** and the incision blade **60** will be described later in detail.

(33) In the exemplary embodiment, the scraper part **24** is composed of a pair of scraper members **24a** and **24b** disposed along the conveyance path of the fixing jig **20** formed by the chain conveyor **12** on opposite sides of the center line **C** (see FIG. **3**) passing through the center of the fixing jig **20**. The scraper members **24a** and **24b** are formed, for example, in a long plate shape, and are arranged in such a way that the distance between them gradually decreases toward the downstream side in the conveying direction of the workpiece **w** (direction of arrow **a**).

(34) When the scraper members **24a** and **24b** of such a shape descend, they contact the surface of scapula bones of the workpiece **w** and can separate the meat part attached to the surface of scapula bones from the scapula bones with good yield.

(35) In the illustrated embodiment, as shown in FIG. 2, a reinforcement bar **48** is attached adjacent to the pair of scraper members **24a** and **24b**.

(36) FIG. 4A is a perspective view of the pair of scraper members **24a** and **24b**.

(37) FIG. 4B is a perspective view of the pair of scraper members **24a** and **24b** when viewed from a different direction than FIG. 4A.

(38) In the illustrated embodiment, one scraper member **24a** of the pair of scraper members **24a** and **24b** has a scraper member body portion **25a** formed in a flat long plate shape, and a back plate portion **25c** extending from the end of the scraper member body portion **25a** on the downstream side in the conveying direction of the workpiece **w** (direction of arrow **a**) toward the other scraper member **24b** of the pair of scraper members **24a** and **24b**.

(39) The other scraper member **24b** of the pair of scraper members **24a** and **24b** has a scraper member body portion **25b** formed in a flat long plate shape.

(40) As well shown in FIGS. 4A and 4B, the lower end surfaces **251a** and **251b** of the pair of scraper member body portions **25a** and **25b** are formed so as to extend downstream in the conveying direction of the workpiece **w** (direction of arrow **a**) as they extend upward.

(41) The lower end surfaces **251a** and **251b** of the pair of scraper member body portions **25a** and **25b** are located on the downstream side in the traveling direction (direction of arrow **c**) in which the scraper part **24** travels toward the fixing jig **20** when the meat part is separated from the scapula bones.

(42) As described below, when the pair of scraper member body portions **25a** and **25b** penetrates the workpiece **w**, the lower end surfaces **251a** and **251b** of the pair of scraper member body portions **25a** and **25b** enter the workpiece **w** from the end portions **252a** and **252b** on the upstream side in the conveying direction of the workpiece **w** (direction of arrow **a**). Therefore, if the end surfaces **251a** and **251b** are formed so as to extend downstream in the conveying direction of the workpiece **w** (direction of arrow **a**) as they extend upward, the pair of scraper member body portions **25a** and **25b** can easily penetrate the workpiece **w**. This makes it easier for the pair of scraper members to reach the surface of scapula bones in separating the meat part from the scapula bones, thus improving the yield of the separated meat part.

(43) As well shown in FIGS. 4A and 4B, the pair of scraper member body portions **25a** and **25b** has cutting edges **253a** and **253b** formed only in a partial region **R1** of the lower end surfaces **251a** and **251b**, including end portions **252a** and **252b** on the upstream side in the conveying direction of the workpiece **w** (direction of arrow **a**).

(44) As described above, when the pair of scraper member body portions **25a** and **25b** penetrates the workpiece **w**, the lower end surfaces **251a** and **251b** enter the workpiece **w** from the end portions **252a** and **252b** on the upstream side in the conveying direction of the workpiece **w** (direction of arrow **a**). Accordingly, the cutting edges **253a** and **253b** formed at the end portions **252a** and **252b** can cut the meat part, so that the pair of scraper member body portions **25a** and **25b** can easily penetrate the workpiece **w**. This makes it easier for the pair of scraper member body portions **25a** and **25b** to reach the surface of scapula bones in separating the meat part from the scapula bones, thus improving the yield of the separated meat part.

(45) In addition, since the cutting edges **253a** and **253b** are not formed in a region **R2** other than the partial region **R1** on the lower end surfaces **251a** and **251b**, the risk of scraping off the scapula bones by the region **R2** is reduced. This reduces the risk of the scapula bones being scraped off and adhering to the meat part as foreign substances.

(46) Attached to the back plate portion **25c** is an incision blade **60** for incising a skin between the scapula bones and a meat guide part **70**.

(47) FIG. 5 is a schematic cross-sectional view showing the separation of a meat part **w_m** around scapula bones **w_b** from the scapula bones **w_b** and the incision of a skin **w_s** between the scapula bones **w_b** with the incision making part **22**.

(48) In the illustrated embodiment, the incision blade **60** has a cutting edge **61** extending in a

direction away from the fixing jig **20**, i.e., downstream in the conveying direction of the workpiece **w** (direction of arrow **a**) as it extends upward, i.e., upstream in a traveling direction (direction of arrow **c**) in which the scraper part **24** travels toward the fixing jig **20** in separating the meat part **w_m** from the scapula bones **w_b**.

(49) In the illustrated embodiment, the meat guide part **70** is disposed on the opposite side of the scraper part **24** from the incision blade **60** and has a plate-like member **71** arranged to overlap the scraper part **24** when viewed from downstream in the conveying direction of the workpiece **w** (direction of arrow **a**). The member **71** is arranged so that at least a part of the member **71** is located upstream of the incision blade **60** in the traveling direction (direction of arrow **c**) when viewed from downstream in the conveying direction of the workpiece **w** (direction of arrow **a**). As well shown in FIG. **4B**, the member **71** is arranged so that at least a part of the member **71** is located on one and the other sides of the incision blade **60** in the thickness direction of the incision blade **60** (direction of arrow **d**).

(50) By lowering the scraper part **24** and the incision blade **60** attached to the scraper part **24** and moving the scraper part **24** along the surface of the scapula bones **w_b** of the workpiece **w**, the pair of scraper member body portions **25a** and **25b** separates the meat part **w_m** with skin from the scapula bones **w_b**, as shown in FIG. **5**.

(51) The meat part **w_m** with skin separated from the scapula bones **w_b** rides up on the pair of scraper member body portions **25a** and **25b**, is guided between the pair of scraper member body portions **25a** and **25b** and the meat guide part **70**, and is cut by the incision blade **60**.

(52) The meat part **w_m** separated from the scapula bones **w_b** rides up on the pair of scraper member body portions **25a** and **25b**, is guided between the pair of scraper member body portions **25a** and **25b** and the meat guide part **70**, and is cut by the incision blade **60**.

(53) With the above-described poultry processing apparatus **10**, the skin **w_s** between the scapula bones **w_b** of the meat part **w_m** that rides up on the scraper part **24** can be cut from the inside of the workpiece **w** by the incision blade **60**. This makes it more difficult to damage the meat of the neck part **w_n** of the workpiece **w**, improving the yield and quality of the meat of the neck part **w_n** of the workpiece **w**.

(54) Additionally, with the above-described poultry processing apparatus **10**, the process of separating the meat part **w_m** around the scapula bones **w_b** from the scapula bones **w_b** and the process of incising the skin **w_s** between the scapula bones **w_b**, which were conventionally performed in two processes, can be performed in a single process, thus improving the processing efficiency of the workpiece **w**.

(55) In the above-described poultry processing apparatus **10**, the incision blade **60** is fixed to the scraper part **24**.

(56) As a result, the skin **w_s** between the scapula bones **w_b** of the meat part **w_m** that rides up on the scraper part **24** can be cut easily.

(57) In the above-described poultry processing apparatus **10**, the scraper part **24** includes a pair of scraper members **24a** and **24b** spaced apart from each other in a direction intersecting the traveling direction (direction of arrow **c**). The incision blade **60** is fixed to one of the pair of scraper members **24a** and **24b** (in the illustrated embodiment, scraper member **24a**).

(58) This allows the pair of scraper members **24a** and **24b** to be aligned with the position and orientation of the scapula bones **w_b** of the workpiece **w**, so that the pair of scraper members **24a** and **24b** can be made to follow the surface of the scapula bones **w_b** reliably when the meat part **w_m** is separated from the scapula bones **w_b**. Thus, it is possible to improve the yield of the separated meat part **w_m**.

(59) In the above-described poultry processing apparatus **10**, the meat part **w_m** with the skin **w_s** can be guided to the incision blade **60** by the meat guide part **70** when the meat part **w_m** is separated from the scapula bones **w_b**, improving the certainty of incision of the skin **w_s** between the scapula bones **w_b** by the incision blade **60**.

(60) (Posture Adjustment Device **40**)

(61) FIG. **6** is a cross-sectional view taken along line B-B in FIG. **2**.

(62) In the illustrated embodiment, the poultry processing apparatus **10** is equipped with a posture adjustment device **40** disposed upstream of the scraper part **24** in the conveying direction of the workpiece **w** (direction of arrow **a**) and configured to raise the neck part **wn** of the workpiece **w**. The posture adjustment device **40** includes a guide member **41** disposed above a track along which the fixing jig **20** moves and configured to guide the neck part **wn** of the workpiece **w**, and a tapered passage **42** disposed above the track and downstream of the guide member **41** in the conveying direction (direction of arrow **a**) and having a passage width **Pw** gradually decreasing toward the downstream side in the conveying direction.

(63) That is, the posture adjustment device **40** for a poultry carcass includes a fixing jig **20**, which is a holder for holding the poultry carcass (workpiece **w**) from the inside while being inserted inside the poultry carcass (workpiece **w**), a guide member **41** disposed above a track along which the fixing jig **20** moves and configured to guide the neck part **wn** of the poultry carcass (workpiece **w**), and a tapered passage **42** disposed above the track and downstream of the guide member **41** in the moving direction of the fixing jig **20** and having a passage width **Pw** gradually decreasing toward the downstream side in the moving direction.

(64) In the illustrated embodiment, the guide member **41** has a pair of first plate members **41a** and **41b** spaced apart from each other in a direction perpendicular to the conveying direction of the workpiece **w** (direction of arrow **a**). The pair of first plate members **41a** and **41b** is arranged so that the distance between the first plate members **41a** and **41b** decreases from upstream to downstream in the conveying direction of the workpiece **w**.

(65) A pair of second plate portions **42a** and **42b** is connected to the downstream side of the pair of first plate members **41a** and **41b** in the conveying direction, respectively, to form the tapered passage **42**.

(66) In the illustrated embodiment, each of the second plate portions **42a** and **42b** extends parallel to the conveying direction (direction of arrow **a**) downstream in the conveying direction. In the illustrated embodiment, a protruding plate portion **43** is attached to one second plate portion **42a** of the pair of second plate portions **42a** and **42b**. The protruding plate portion **43** is formed so that the distance between the protruding plate portion **43** and the other second plate portion **42b** of the pair of second plate portions **42a** and **42b** decreases from upstream to downstream in the conveying direction. Thus, the tapered passage **42** is formed by the protruding plate portion **43** and the other second plate portion **42b** so that the passage width **Pw** decreases toward the downstream side in the conveying direction.

(67) The protruding plate portion **43** and the other second plate portion **42b** are a pair of wall portions that constitute the tapered passage **42**.

(68) In the illustrated embodiment, the relative position between the protruding plate portion **43** and the other second plate portion **42b** is invariable.

(69) In the illustrated embodiment, one of the pair of wall portions and the other, the protruding plate portion **43** and the other second plate portion **42b**, are asymmetrically formed with respect to a center line **C** through which the center position of the fixing jig **20** passes when viewed from above, i.e., the center line **C** passing through the center of the fixing jig **20**. The protruding plate portion **43**, one of the pair of wall portions, is formed so as to gradually approach the other second plate portion **42b**, the other of the pair of wall portions, toward the downstream side in the moving direction.

(70) FIGS. **7A**, **7B**, **7C**, and **7D** are diagrams for describing the operation of the posture adjustment device **40** to raise the neck part **wn** that has fallen down to the downstream side in the moving direction.

(71) The workpiece **w** mounted on the fixing jig **20** has its back side facing downstream in the moving direction so that the neck part **wn** falls down to the downstream side in the moving

direction. Therefore, the fallen neck part wn may interfere with the separation of the meat part wm around the scapula bones wb and the incision of the skin ws between the scapula bones wb, as described above.

(72) For this reason, in the poultry processing apparatus **10** according to some embodiments, the posture adjustment device **40** is intended to raise the neck part wn just before the separation of the meat part wm around the scapula bones wb and the incision of the skin ws between the scapula bones wb.

(73) The workpiece w mounted on the fixing jig **20** moves downstream in the moving direction as shown in FIG. 7A, and when it reaches the posture adjustment device **40** as shown in FIG. 7B, the root portion nb of the neck part wn is guided by the guide member **41** to the tapered passage **42**.

(74) In the illustrated embodiment, when the neck part wn passes through the tapered passage **42**, since the passage width Pw becomes narrower toward the downstream side in the moving direction so that the neck part wn gradually becomes difficult to pass through the tapered passage **42**, the neck part wn that has fallen toward the downstream side in the moving direction can be raised toward the upstream side in the moving direction.

(75) More specifically, as shown in FIG. 7C, when the root portion nb of the neck part wn is caught in the tapered passage **42**, the portion of the workpiece w below the root portion nb moves downstream of the root portion nb in the moving direction, causing the neck part wn to rotate clockwise (direction of arrow e) around the root portion nb. As a result, the neck part wn on the tip side than the root portion nb rotates clockwise and rises as shown in FIG. 7D.

(76) After the root portion nb passes through the tapered passage **42**, the neck part wn tries to fall down again toward the downstream side in the moving direction. However, the neck part wn is prevented from falling down by the scraper part **24** or the bracket **32**, which are located downstream of the posture adjustment device **40** in the moving direction, as they come into contact with the neck part wn.

(77) In the above-described poultry processing apparatus **10**, by raising the neck part wn of the workpiece w by the posture adjustment device **40** just before separating the meat part wm from the scapula bones wb, the neck part wn is not in the way when the meat part wm is separated from the scapula bones wb, improving the certainty of separation of the meat part wm from the scapula bones wb.

(78) In the above-described poultry processing apparatus **10**, the posture adjustment device **40** may move upstream in the moving direction relative to the fixing jig **20** to raise the neck part wn. Specifically, in the above-described poultry processing apparatus **10**, the posture adjustment device **40** may be immobile while the fixing jig **20** may move downstream in the moving direction, or the posture adjustment device **40** itself may move upstream in the moving direction. In the above-described poultry processing apparatus **10**, the neck part wn can be raised by relative movement of the posture adjustment device **40** and the fixing jig **20** in the moving direction.

(79) Thus, the neck part wn can be easily raised by the posture adjustment device **40** with a simple configuration.

(80) In the above-described posture adjustment device **40**, since the relative position between the protruding plate portion **43** and the other second plate portion **42b** is invariable, it is not necessary to move the protruding plate portion **43** and the other second plate portion **42b** relative to each other, and the configuration of the posture adjustment device **40** can be simplified.

(81) Additionally, in the above-described posture adjustment device **40**, since the relative position between the tapered passage **42** and the guide member **41** is invariable, it is not necessary to move the tapered passage **42** and the guide member **41** relative to each other, and the configuration of the posture adjustment device **40** can be simplified.

(82) In the above-described posture adjustment device **40**, as previously described, since the protruding plate portion **43** and the other second plate portion **42b** are asymmetrically formed with respect to the center line C passing through the center of the fixing jig **20**, even with the same

passage width **Pw**, the resistance of the neck part **wn** to pass through the tapered passage **42** is greater than when they are symmetrically formed.

(83) The neck part **wn** can be raised more reliably by increasing the resistance of the neck part **wn** as it passes through the tapered passage **42**. However, if the passage width **Pw** of the tapered passage **42** is narrowed to increase the resistance of the neck part **wn** as it passes through the tapered passage **42**, since the thickness of the neck part **wn** varies for each poultry carcass, the resistance may be excessively high for some poultry carcasses, unnecessarily increasing the burden on the poultry carcass. Therefore, if the protruding plate portion **43** and the other second plate portion **42b** are asymmetrically formed with respect to the center line **C** when viewed from above, the resistance of the neck part **wn** to pass through the tapered passage **42** can be increased while maintaining some passage width **Pw**. Thus, the burden on the poultry carcass when the neck part **wn** passes through the tapered passage **42** can be suppressed.

(84) The present disclosure is not limited to the embodiments described above, but includes modifications to the embodiments described above, and embodiments composed of combinations of those embodiments.

(85) For example, in the above-described embodiments, the posture adjustment device **40** is used to raise the neck part **wn** just before separating the meat part **wm** around the scapula bones **wb** and incising the skin **ws** between the scapula bones **wb**. However, the use of the posture adjustment device **40** is not limited to the use described above and may be used for other processes in the poultry processing.

(86) The contents described in the above embodiments would be understood as follows, for instance.

(87) (1) A poultry processing apparatus **10** according to at least one embodiment of the present disclosure includes: a scraper part **24** for separating a meat part **wm** around scapula bones **wb** of a poultry carcass (workpiece **w**) held by a holder (fixing jig **20**) from the scapula bones **wb**; a cutting edge extending in a direction away from the holder (fixing jig **20**) as the cutting edge extends upstream in a traveling direction in which the scraper part **24** travels toward the holder (fixing jig **20**) in separating the meat part **wm** from the scapula bones **wb**; and an incision blade **60** with the cutting edge **61** for incising a skin **ws** between the scapula bones **wb** of the meat part **wm** that rides up on the scraper part **24**.

(88) With the above configuration (1), the skin **ws** between the scapula bones **wb** of the meat part **wm** that rides up on the scraper part **24** can be cut from the inside of the poultry carcass (workpiece **w**) by the incision blade **60**. This makes it more difficult to damage the meat of the neck part **wn** of the poultry carcass (workpiece **w**), improving the yield and quality of the meat of the neck part **wn** of the poultry carcass (workpiece **w**).

(89) Additionally, with the above configuration (1), the process of separating the meat part **wm** around the scapula bones **wb** from the scapula bones **wb** and the process of incising the skin **ws** between the scapula bones **wb**, which were conventionally performed in two processes, can be performed in a single process, thus improving the processing efficiency of the poultry carcass.

(90) (2) In some embodiments, in the above configuration (1), the incision blade **60** may be fixed to the scraper part **24**.

(91) With the above configuration (2), the skin **ws** between the scapula bones **wb** of the meat part **wm** that rides up on the scraper part **24** can be cut easily.

(92) (3) In some embodiments, in the above configuration (1) or (2), the scraper part **24** may include a pair of scraper members **24a** and **24b** spaced apart from each other in a direction intersecting the traveling direction. The incision blade **60** may be fixed to one of the pair of scraper members **24a** and **24b**.

(93) With the above configuration (3), the pair of scraper members **24a** and **24b** can be aligned with the position and orientation of the scapula bones **wb** of the poultry carcass (workpiece **w**), so that the pair of scraper members **24a** and **24b** can be made to follow the surface of the scapula bones

wb reliably when the meat part wm is separated from the scapula bones wb. Thus, it is possible to improve the yield of the separated meat part wm.

(94) Additionally with the above configuration (3), the incision blade **60** can be easily fixed to one of the pair of scraper members **24a** and **24b**.

(95) (4) In some embodiments, in the above configuration (3), end surfaces **251a** and **251b** of the pair of scraper members **24a** and **24b** on a downstream side in the traveling direction may extend downstream in a moving direction of the holder (fixing jig **20**) as the end surfaces extend upstream in the traveling direction.

(96) With the above configuration (4), when the pair of scraper members **24a** and **24b** penetrates the poultry carcass (workpiece w), the end surfaces (end surfaces **251a** and **251b**) of the pair of scraper members **24a** and **24b** on the downstream side in the traveling direction enter the poultry carcass from the end portions **252a** and **252b** on the upstream side in the moving direction of the holder (fixing jig **20**), so that the pair of scraper members **24a** and **24b** can easily penetrate the poultry carcass (workpiece w). This makes it easier for the pair of scraper members **24a** and **24b** to reach the surface of scapula bones wb in separating the meat part wm from the scapula bones wb, thus improving the yield of the separated meat part wm.

(97) (5) In some embodiments, in the above configuration (4), the pair of scraper members **24a** and **24b** may have cutting edges **253a** and **253b** formed only in a partial region R1 of the end surfaces (end surfaces **251a** and **251b**) of the scraper members **24a** and **24b** on the downstream side in the traveling direction, the partial region R1 including end portions **252a** and **252b** on the upstream side in the moving direction of the holder (fixing jig **20**).

(98) With the above configuration (5), the pair of scraper members **24a** and **24b** can easily reach the surface of scapula bones wb in separating the meat part wm from the scapula bones wb, thus improving the yield of the separated meat part wm.

(99) Additionally, with the above configuration (5), since the cutting edges **253a** and **253b** are not formed in a region R2 other than the partial region R1 on the end surfaces **251a** and **251b** of the pair of scraper members **24a** and **24b** on the downstream side in the traveling direction, the risk of scraping off the scapula bones wb by the region R2 is reduced. This reduces the risk of the scapula bones wb being scraped off and adhering to the meat part wm as foreign substances.

(100) (6) In some embodiments, in any one of the above configurations (1) to (5), the apparatus may include a meat guide part **70** disposed on an opposite side of the scraper part **24** from the incision blade **60**.

(101) With the above configuration (6), the meat part wm with skin ws can be guided to the incision blade **60** by the meat guide part **70** when the meat part wm is separated from the scapula bones wb, improving the certainty of incision of the skin ws between the scapula bones wb by the incision blade **60**.

(102) (7) In some embodiments, in any one of the above configurations (1) to (6), the apparatus may include a posture adjustment device **40** disposed upstream of the scraper part **24** in a moving direction of the holder (fixing jig **20**) and configured to raise a neck wn part of the poultry carcass (workpiece w). The posture adjustment device **40** may include: a guide member **41** disposed above a track along which the holder (fixing jig **20**) moves and configured to guide the neck part wn of the poultry carcass (workpiece w); and a tapered passage **42** disposed above the track and downstream of the guide member **41** in the moving direction of the holder (fixing jig **20**) and having a width Pw gradually decreasing toward a downstream side in the moving direction.

(103) With the above configuration (7), when the neck part wn passes through the tapered passage **42**, since the passage width Pw becomes narrower toward the downstream side in the moving direction so that the neck part wn gradually becomes difficult to pass through the tapered passage **42**, the neck part wn that has fallen toward the downstream side in the moving direction can be raised toward the upstream side in the moving direction. By raising the neck part wn of the poultry carcass (workpiece w) by the posture adjustment device **40** just before separating the meat part wm

from the scapula bones **wb**, the neck part **wn** is not in the way when the meat part **wm** is separated from the scapula bones **wb**, improving the certainty of separation of the meat part **wm** from the scapula bones **wb**.

(104) (8) In some embodiments, in the above configuration (7), the posture adjustment device **40** may be configured to move upstream in the moving direction relative to the holder (fixing jig **20**) to raise the neck part **wn**.

(105) With the above configuration (8), the neck part **wn** can be raised easily.

(106) (9) In some embodiments, in the above configuration (7) or (8), the tapered passage **42** may be composed of a pair of wall portions (protruding plate portion **43** and other second plate portion **42b**) spaced apart from each other in a direction intersecting the moving direction. Further, a relative position between one (protruding plate portion **43**) of the pair of wall portions (protruding plate portion **43** and other second plate portion **42b**) and the other (other second plate portion **42b**) may be invariable.

(107) With the above configuration (9), since it is not necessary to move one (protruding plate portion **43**) of the pair of wall portions (protruding plate portion **43** and other second plate portion **42b**) and the other (other second plate portion **42b**) relative to each other, the configuration of the posture adjustment device **40** can be simplified.

(108) (10) In some embodiments, in the above configuration (9), one (protruding plate portion **43**) of the pair of wall portions (protruding plate portion **43** and other second plate portion **42b**) and the other (other second plate portion **42b**) may be asymmetrically formed with respect to a center line (center line C) through which a center position of the holder (fixing jig **20**) passes when viewed from above, and one (protruding plate portion **43**) of the pair of wall portions may be formed so as to gradually approach the other (other second plate portion **42b**) of the pair of wall portions toward a downstream side in the moving direction of the holder (fixing jig **20**).

(109) With the above configuration (10), the resistance of the neck part **wn** to pass through the tapered passage **42** can be increased while maintaining some passage width **Pw**. Thus, the burden on the poultry carcass (workpiece **w**) when the neck part **wn** passes through the tapered passage **42** can be suppressed.

(110) (11) In some embodiments, in any one of the above configurations (7) to (10), a relative position between the tapered passage **42** and the guide member **41** may be invariable.

(111) With the above configuration (11), it is not necessary to move the tapered passage **42** and the guide member **41** relative to each other, and the configuration of the posture adjustment device **40** can be simplified.

REFERENCE SIGNS LIST

(112) **10** Poultry processing apparatus **20** Fixing jig (Holder) **22** Incision making part **24** Scraper part **24a**, **24b** Scraper member **40** Posture adjustment device **41** Guide member **42** Tapered passage **60** Incision blade **70** Meat guide part

Claims

1. A poultry processing apparatus, comprising: a scraper part for separating a meat part around scapula bones of a poultry carcass held by a holder from the scapula bones; a cutting edge extending in a direction away from the holder as the cutting edge extends upstream in a traveling direction in which the scraper part travels toward the holder in separating the meat part from the scapula bones, and an incision blade with the cutting edge for incising a skin between the scapula bones of the meat part that rides up on the scraper part.
2. The poultry processing apparatus according to claim 1, wherein the incision blade is fixed to the scraper part.
3. The poultry processing apparatus according to claim 1, wherein the scraper part includes a pair of scraper members spaced apart from each other in a direction intersecting the traveling direction,

and wherein the incision blade is fixed to one of the pair of scraper members.

4. The poultry processing apparatus according to claim 3, wherein end surfaces of the pair of scraper members on a downstream side in the traveling direction extend downstream in a moving direction of the holder as the end surfaces extend upstream in the traveling direction.

5. The poultry processing apparatus according to claim 4, wherein the pair of scraper members has cutting edges formed only in a partial region of the end surfaces of the scraper members on the downstream side in the traveling direction, the partial region including upstream end portions of the end surfaces in the moving direction of the holder.

6. The poultry processing apparatus according to claim 1, comprising a meat guide part disposed on an opposite side of the scraper part from the incision blade.

7. The poultry processing apparatus according to claim 1, comprising a posture adjustment device disposed upstream of the scraper part in a moving direction of the holder and configured to raise a neck part of the poultry carcass, wherein the posture adjustment device includes: a guide member disposed above a track along which the holder moves and configured to guide the neck part of the poultry carcass; and a tapered passage disposed above the track and downstream of the guide member in the moving direction of the holder and having a width gradually decreasing toward a downstream side in the moving direction.

8. The poultry processing apparatus according to claim 7, wherein the posture adjustment device is configured to move upstream in the moving direction relative to the holder to raise the neck part.

9. The poultry processing apparatus according to claim 7, wherein the tapered passage is composed of a pair of wall portions spaced apart from each other in a direction intersecting the moving direction, and wherein a relative position between one of the pair of wall portions and the other is invariable.

10. The poultry processing apparatus according to claim 9, wherein one of the pair of wall portions and the other are asymmetrically formed with respect to a center line through which a center position of the holder passes when viewed from above, and wherein one of the pair of wall portions is formed so as to gradually approach the other of the pair of wall portions toward a downstream side in the moving direction of the holder.

11. The poultry processing apparatus according to claim 7, wherein a relative position between the tapered passage and the guide member is invariable.
